DISCRETE MATH · SEQUENCES, MATHEMATICAL INDUCTION, AND RECURSION

Recursion & Closed Form Explorer

Explore & Reflect

Name _____ Date _____ Period _____

Open **Recursion & Closed Form Explorer** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. $a_n = 2a_{n-1} + 1$ with $a_1 = 1$ (Tower of Hanoi). Predict a_5 and a_{10}.

2. Same rule, but $a_1 = 4$. Do you get the same sequence? Why does a recurrence need initial terms?

2 Explore use the tool

1. Arithmetic: set c = 1. Find d and a_1 so that $a_1 = 5$ and $a_{10} = 32$. Record the closed form.

2. Tower of Hanoi: type a closed form that matches every term. What does the tool say about proof?

3. Find c, d and a_1 with $c \neq 1$ that make a constant sequence. What is the relationship?

4. Fibonacci ($a_1 = a_2 = 1$): record a_{12} and S_{10}. Compare S_{10} with a_{12}.

3 Record use the tool and fill in the table

Recurrences and closed forms

Enter each recurrence, then test your closed form in the guess box.

Recurrence and initial term	$a_1 \ldots a_5$	Type	Closed form
$a_n = a_{n-1} + 4$, $a_1 = 7$			
$a_n = 3a_{n-1}$, $a_1 = 2$			
$a_n = 2a_{n-1} + 1$, $a_1 = 1$			
$a_n = 0.5a_{n-1} + 3$, $a_1 = 2$			

4 Explain what did you discover?

1. If c = 1 the sequence is _____ with common difference d; if d = 0 it is _____ with common ratio c.

2. A closed form that matches many terms is a _____ until it is proved.

3. Induction proof of $a_n = f(n)$: show $f(1) = a_1$ (the _____ step), then assume $a_k = f(k)$ and show $c \cdot f(k) + d =$ _____.

4. Prove by induction that $a_n = 2^n - 1$ for $a_n = 2a_{n-1} + 1$, $a_1 = 1$.

5 Apply a new situation, no tool

1. $a_n = a_{n-1} - 6$, $a_1 = 50$. Find a_{20} with a closed form.

2. A $1000 deposit grows 5% a year and $100 is added each year after the first: $a_n = 1.05a_{n-1} + 100$, $a_1 = 1000$. Find a_3.

3. Find a closed form for $a_n = 3a_{n-1}$, $a_1 = 5$, and find a_6.

4. Show that the guess $a_n = n^2 - n + 1$ matches $a_n = a_{n-1} + 2(n - 1)$, $a_1 = 1$ for $n = 1, 2, 3$, then do the inductive step.